

ADAM MESSAOUDI

Data Engineer | Business Intelligence Engineer | Data Analyst

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PROFESSIONAL SUMMARY

Data Engineer experienced in ETL/ELT pipelines, data modeling, Microsoft Fabric Lakehouse, Power BI dashboards, SQL, Python automation, AI-assisted development, and business intelligence solutions. Skilled in transforming complex datasets into actionable insights and building scalable analytics platforms. Currently refining German language skills (B1 level) and motivated to contribute to international data engineering environments.

PROFESSIONAL EXPERIENCE

End-of-Studies Data Engineering Intern
ZF Group

03/2025 – 09/2025
Hanover, Germany

- Project: Enterprise Data Platform & Predictive Analytics for Global Workshop Networks
- Architected and deployed scalable ETL/ELT data pipelines using Microsoft Fabric (OneLake, Dataflows Gen2, Semantic Models) and Salesforce, integrating 130,000+ records into a centralized Lakehouse environment.
 - Automated public review data extraction across 4,600+ workshops using Python and Selenium, enriching internal CRM records with Google ratings and review metrics through automated data integration workflows.
 - Engineered 5 interactive Power BI dashboards establishing a standardized Workshop Health Score (0–100) combining YTD audit timeliness, Google ratings, review engagement, and internal audit performance indicators.
 - Developed an API-driven NLP Data Assistant using spaCy and OpenAI API, enabling conversational KPI retrieval and natural language interaction with enterprise data.
 - Implemented real-time YTD audit compliance tracking and "Audit Today" operational alerts, providing regional managers with immediate visibility into overdue and upcoming workshop audits.
 - Developed a machine learning forecasting pipeline using MLflow and regression models (Ridge, Random Forest, XGBoost) to evaluate model performance, select the best-performing approach, and predict workshop network evolution across regions.
 - Technologies:** Microsoft Fabric (OneLake, Dataflows Gen2), Power BI (Power Query, DAX), Python (Pandas, Selenium, spaCy, Scikit-Learn), OpenAI API, MLflow, Flask, REST APIs, SQL, Salesforce.

Data Engineering Intern
Infraway Tunisia

06/2024 – 09/2024
Tunis, Tunisia

- Project: Telecom Management Optimization Through Data-Driven Insights
- Analyzed 10.01M+ telecommunications call records across 43 attributes via EDA to uncover key network traffic trends.
 - Streamlined ETL workflows by building automated Talend staging jobs, reducing data processing time by 45% and enabling daily dashboard refreshes.
 - Designed 2 Power BI dashboards (Service Quality & Profitability) for Sales Managers and Service Operators, optimizing 63 B2B accounts.
 - Trained Exponential Smoothing forecasting models in Python (Pandas, Statsmodels) to predict monthly call volumes and support capacity planning.
 - Technologies:** MySQL, Talend, Power BI, Python, Machine Learning (Statsmodels, Scikit-Learn), SQL.

Business Intelligence Intern
PGS International

06/2023 – 09/2023
Ben Arous, Tunisia

- Project: Data-Driven Inventory Optimization Platform for Oil Depots
- Centralized oil depot inventory data across 5 depot locations into a PostgreSQL data warehouse using automated Talend ETL workflows.
 - Visualized real-time stock levels and 10+ operational KPIs across 3 Power BI dashboards to track inventory performance.
 - Orchestrated Python web scrapers (Selenium, BeautifulSoup) parsing market data across 4 target industry sources daily to enrich analytics.
 - Optimized 10+ backend REST API routes of a Flask analytics platform to support data-driven inventory management.
 - Technologies:** Talend, PostgreSQL, Power BI, Python, Angular, Flask, Machine Learning, Web Scraping (Selenium, BeautifulSoup).

TECHNICAL SKILLS

- Data Engineering & BI:** Microsoft Fabric (OneLake, Lakehouse, Dataflows Gen2, Pipelines, Semantic Models), Power BI (Service, Power Query, DAX), ETL/ELT, Data Pipelines, Data Warehousing, Data Modeling, Data Integration, Data Quality, Data Governance, KPI Reporting.
- Programming, Databases & Data Platforms:** Python (Pandas, Flask, Selenium), SQL, PostgreSQL, MySQL, Oracle SQL, Microsoft SQL Server (SSIS, SSMS, SSRS), MongoDB, REST APIs, JSON, CSV, Git, Docker, Linux.
- Machine Learning & AI:** Machine Learning, Predictive Modeling, Statistical Analysis, Time Series Forecasting, Scikit-Learn, MLflow, NLP (spaCy), OpenAI API, Azure OpenAI, Generative AI Concepts.
- Enterprise Applications & Cloud:** Salesforce, Microsoft Dynamics 365, Azure (Azure Storage, Azure SQL, Azure Entra ID), SAP BW.
- Data Automation & Methodologies:** Web Scraping (Selenium, BeautifulSoup), Data Extraction, Data Cleaning, Data Enrichment, Workflow Automation, Agile/Scrum, Requirements Analysis.

EDUCATION

ESPRIT School of Engineering – Tunisia
Engineering Degree in Computer Science (Master's Equivalent)

Graduated: 10/2025

Schmalkalden University of Applied Sciences – Germany
Exchange Program in Computer Science

10/2024 – 09/2025

Borj Cedria High School – Tunisia
Baccalaureate in Experimental Sciences

08/2020

CERTIFICATIONS

SAP Solution Implementation – SAP | Coursera

11/2024

Microsoft Azure Cloud Services – Microsoft | Coursera

10/2024

MLOps (Machine Learning Operations) – Duke University | Coursera

10/2024

Talend Data Integration – Talend | Coursera

10/2024

LANGUAGES

German: B1 (In Progress) | English: Fluent | French: Fluent | Spanish: Elementary | Arabic: Native